ITI – RedHat Admin 2 Lab 1

Mohammad Tarek Omar – Data Management

1. Use systemctl to view the status of all the system services.

```
mohammadtarek@localhost:~ — systemctl list-units --type...

[mohammadtarek@localhost ~]$
[mohammadtarek@localhost ~]$ ^C
[mohammadtarek@localhost ~]$ systemctl list-units --type=service
  UNIT                                  LOAD   ACTIVE SUB     DESCRIPTION       >
  accounts-daemon.service               loaded active running Accounts Service
  alsa-state.service                    loaded active running Manage Sound Card>
  atd.service                           loaded active running Deferred executio>
  auditd.service                        loaded active running Security Auditing>
  avahi-daemon.service                  loaded active running Avahi mDNS/DNS-SD>
  chronyd.service                       loaded active running NTP client/server
  colord.service                        loaded active running Manage, Install a>
  crond.service                         loaded active running Command Scheduler
  cups.service                          loaded active running CUPS Scheduler
  dbus-broker.service                   loaded active running D-Bus System Mess>
  dracut-shutdown.service               loaded active exited  Restore /run/init>
  firewalld.service                     loaded active running firewalld - dynam>
  fwupd.service                         loaded active running Firmware update d>
  gdm.service                           loaded active running GNOME Display Man>
  kdump.service                         loaded active exited  Crash recovery ke>
  kmod-static-nodes.service             loaded active exited  Create List of St>
  libstoragemgmt.service                loaded active running libstoragemgmt pl>
  lvm2-monitor.service                  loaded active exited  Monitoring of LVM>
  mcelog.service                        loaded active running Machine Check Exc>
  ModemManager.service                  loaded active running Modem Manager
  NetworkManager-wait-online.service loaded active exited  Network Manager W>
  NetworkManager.service                loaded active running Network Manager
```

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2. change the default target back to multi-user.target and reboot:

```
mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ sudo systemctl set-default multi-user.target
[sudo] password for mohammadtarek:
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/
multi-user.target.
[mohammadtarek@localhost ~]$ rebot
bash: rebot: command not found...
rebooInstall package 'python3-robotframework' to provide command 'rebot'? [N/
y] ^C
[mohammadtarek@localhost ~]$ reboot
```

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3. Send mail to the root user.

4.Verify that you have received this mail.

```
[mohammadtarek@localhost ~]$ echo "test redhat admin 2 lab 1" | mail -s "Mail te
st" mohammadtarek
You have new mail in /var/spool/mail/mohammadtarek
[mohammadtarek@localhost ~]$ mail
s-nail version v14.9.22.  Type `?' for help
/var/spool/mail/mohammadtarek: 1 message 1 new
▸N  1 Mohammad Tarek        2026-01-17 14:39   19/860   "Mail test            "
&
```

5.Use systemctl utility to stop postfix/sendmail service.

```
mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ sudo systemctl stop postfix
[mohammadtarek@localhost ~]$ sudo systemctl stop sendmail
```

6.Send mail again to the root user.

7.Verify that you have received this mail.

```
Sat 17 Jan 14:45:45 EET 2026
[mohammadtarek@localhost ~]$ mail "test" root
Subject: asd
To: test, root
asdasd


asd
^D
-------
(Preliminary) Envelope contains:
To: test, root
Subject: asd
Send this message [yes/no, empty: recompose]? y
[mohammadtarek@localhost ~]$ mail
s-nail version v14.9.22.  Type `?' for help
/var/spool/mail/mohammadtarek: 1 message
▸   1 Mohammad Tarek        2026-01-17 14:39   20/871   "Mail test
&
```

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8. Use systemctl utility to start postfix/sendmail service

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9. Verify that you have received this mail.

```
^D
-------
(Preliminary) Envelope contains:
To: test3, root
Subject: asd
Send this message [yes/no, empty: recompose]? y
[mohammadtarek@localhost ~]$ mail
s-nail version v14.9.22.  Type `?' for help
/var/spool/mail/mohammadtarek: 1 message
▸   1 Mohammad Tarek        2026-01-17 14:39   20/871   "Mail test
& ^D
Held 1 message in /var/spool/mail/mohammadtarek
[mohammadtarek@localhost ~]$ sudo su -
[root@localhost ~]# mail
s-nail version v14.9.22.  Type `?' for help
/var/spool/mail/root: 5 messages 5 unread
▸U  1 Mohammad Tarek        2026-01-17 14:35   19/815   "Mail test
 U  2 Mohammad Tarek        2026-01-17 14:36   20/825   "Mail test
 U  3 Mohammad Tarek        2026-01-17 14:37   20/825   "Mail test
 U  4 Mohammad Tarek        2026-01-17 14:37   20/825   "Mail test
 U  5 Mohammad Tarek        2026-01-17 14:38   20/825   "Mail test
&
```

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10. switch to the multi-user target manually without rebooting

```
mohammadtarek@localhost:~
[root@localhost ~]# sudo su - mohammadtarek
Sat 17 Jan 14:49:34 EET 2026
[mohammadtarek@localhost ~]$ sudo systemctl set-default multi-user.target
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/mul
ti-user.target.
[mohammadtarek@localhost ~]$
```

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11. configure your server to automatically boot into the multi-user target

- **12. reboot the server, then from within the boot loader menu, boot into rescue.target**

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13. set the default systemd target back to graphical.target

```
[mohammadtarek@localhost ~]$ sudo systemctl set-default graphical.target
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/gra
phical.target.
[mohammadtarek@localhost ~]$
```

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14. Display the status of sshd service, note the PID of the daemon

```
[mohammadtarek@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Sat 2026-01-17 14:16:26 EET; 37min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 811 (sshd)
      Tasks: 1 (limit: 10519)
     Memory: 608.0K (peak: 2.8M)
        CPU: 42ms
     CGroup: /system.slice/sshd.service
             └─811 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 17 14:16:26 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jan 17 14:16:26 localhost.localdomain sshd[811]: Server listening on 0.0.0.0 po>
Jan 17 14:16:26 localhost.localdomain sshd[811]: Server listening on :: port 22.
Jan 17 14:16:26 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)
```

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15. Restart the sshd service and view the status, The PID of the daemon has changed

```
[mohammadtarek@localhost ~]$ sudo systemctl restart sshd
[mohammadtarek@localhost ~]$
```

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16. Reload the sshd service and view the status, The PID of the daemon has not changed and
connection has not be interrupted

```
[mohammadtarek@localhost ~]$ sudo systemctl reload  sshd
[mohammadtarek@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Sat 2026-01-17 14:54:44 EET; 49s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 33780 ExecReload=/bin/kill -HUP $MAINPID (code=exited, status=0/SU>
   Main PID: 33771 (sshd)
      Tasks: 1 (limit: 10519)
     Memory: 1.5M (peak: 1.9M)
        CPU: 58ms
     CGroup: /system.slice/sshd.service
             └─33771 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 17 14:54:44 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jan 17 14:54:44 localhost.localdomain sshd[33771]: Server listening on 0.0.0.0 >
Jan 17 14:54:44 localhost.localdomain sshd[33771]: Server listening on :: port >
Jan 17 14:54:44 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
Jan 17 14:55:29 localhost.localdomain systemd[1]: Reloading OpenSSH server daem>
Jan 17 14:55:29 localhost.localdomain sshd[33771]: Received SIGHUP; restarting.
Jan 17 14:55:30 localhost.localdomain sshd[33771]: Server listening on 0.0.0.0 >
```

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17. Verify that the chronyd service is running

```
[mohammadtarek@localhost ~]$ systemctl status chronyd
● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: >
     Active: active (running) since Sat 2026-01-17 14:16:20 EET; 40min ago
       Docs: man:chronyd(8)
             man:chrony.conf(5)
   Main PID: 734 (chronyd)
      Tasks: 1 (limit: 10519)
     Memory: 952.0K (peak: 4.4M)
        CPU: 196ms
     CGroup: /system.slice/chronyd.service
             └─734 /usr/sbin/chronyd -F 2

Jan 17 14:16:20 localhost chronyd[734]: Loaded 0 symmetric keys
Jan 17 14:16:20 localhost chronyd[734]: Using right/UTC timezone to obtain leap>
Jan 17 14:16:20 localhost chronyd[734]: Frequency -8.099 +/- 1.050 ppm read fro>
Jan 17 14:16:20 localhost chronyd[734]: Loaded seccomp filter (level 2)
Jan 17 14:16:20 localhost systemd[1]: Started NTP client/server.
Jan 17 14:16:38 localhost.localdomain chronyd[734]: Selected source 82.64.247.1>
Jan 17 14:16:38 localhost.localdomain chronyd[734]: System clock wrong by 1.690>
Jan 17 14:16:40 localhost.localdomain chronyd[734]: System clock was stepped by>
Jan 17 14:16:40 localhost.localdomain chronyd[734]: System clock TAI offset set>
Jan 17 14:16:41 localhost.localdomain chronyd[734]: Selected source 109.190.177>
lines 1-22/22 (END)
```

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18. Stop the service and view the status

```
mohammadtarek@localhost:~

[mohammadtarek@localhost ~]$ systemctl stop chronyd
[mohammadtarek@localhost ~]$ systemctl status chronyd
○ chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: >
     Active: inactive (dead) since Sat 2026-01-17 14:58:20 EET; 3min 45s ago
   Duration: 41min 59.918s
       Docs: man:chronyd(8)
             man:chrony.conf(5)
   Main PID: 734 (code=exited, status=0/SUCCESS)
        CPU: 206ms

Jan 17 14:16:20 localhost systemd[1]: Started NTP client/server.
Jan 17 14:16:38 localhost.localdomain chronyd[734]: Selected source 82.64.247.1>
```

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19. Determine if the chronyd service is enabled to start at the system boot

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20. Reboot the system, then view the status of the chronyd service

```
mohammadtarek@localhost:~ — sudo systemctl status chronyd

● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; enabled; preset: >
     Active: active (running) since Sat 2026-01-17 15:03:57 EET; 2min 50s ago
       Docs: man:chronyd(8)
             man:chrony.conf(5)
    Process: 722 ExecStart=/usr/sbin/chronyd $OPTIONS (code=exited, status=0/SU>
   Main PID: 758 (chronyd)
      Tasks: 1 (limit: 10519)
     Memory: 4.2M (peak: 4.4M)
        CPU: 108ms
     CGroup: /system.slice/chronyd.service
             └─758 /usr/sbin/chronyd -F 2

Jan 17 15:03:56 localhost systemd[1]: Starting NTP client/server...
Jan 17 15:03:57 localhost chronyd[758]: chronyd version 4.8 starting (+CMDMON +>
Jan 17 15:03:57 localhost chronyd[758]: Loaded 0 symmetric keys
Jan 17 15:03:57 localhost chronyd[758]: Using right/UTC timezone to obtain leap>
Jan 17 15:03:57 localhost chronyd[758]: Frequency -8.967 +/- 0.982 ppm read fro>
Jan 17 15:03:57 localhost chronyd[758]: Loaded seccomp filter (level 2)
Jan 17 15:03:57 localhost systemd[1]: Started NTP client/server.
Jan 17 15:04:16 localhost.localdomain chronyd[758]: Selected source 5.39.80.51 >
Jan 17 15:04:16 localhost.localdomain chronyd[758]: System clock TAI offset set>
Jan 17 15:05:22 localhost.localdomain chronyd[758]: Selected source 31.58.102.1>
lines 1-23
```

21.Disable the chronyd service so that it doesn't start at system boot, then view the status of

the service

```
mohammadtarek@localhost:~ — sudo systemctl status chronyd

[mohammadtarek@localhost ~]$ sudo systemctl status chronyd
● chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; disabled; preset:>
     Active: active (running) since Sat 2026-01-17 15:03:57 EET; 4min 36s ago
       Docs: man:chronyd(8)
             man:chrony.conf(5)
   Main PID: 758 (chronyd)
      Tasks: 1 (limit: 10519)
     Memory: 4.2M (peak: 4.4M)
        CPU: 111ms
     CGroup: /system.slice/chronyd.service
             └─758 /usr/sbin/chronyd -F 2

Jan 17 15:03:56 localhost systemd[1]: Starting NTP client/server...
Jan 17 15:03:57 localhost chronyd[758]: chronyd version 4.8 starting (+CMDMON +>
Jan 17 15:03:57 localhost chronyd[758]: Loaded 0 symmetric keys
Jan 17 15:03:57 localhost chronyd[758]: Using right/UTC timezone to obtain leap>
Jan 17 15:03:57 localhost chronyd[758]: Frequency -8.967 +/- 0.982 ppm read fro>
Jan 17 15:03:57 localhost chronyd[758]: Loaded seccomp filter (level 2)
Jan 17 15:03:57 localhost systemd[1]: Started NTP client/server.
Jan 17 15:04:16 localhost.localdomain chronyd[758]: Selected source 5.39.80.51 >
Jan 17 15:04:16 localhost.localdomain chronyd[758]: System clock TAI offset set>
Jan 17 15:05:22 localhost.localdomain chronyd[758]: Selected source 31.58.102.1>
lines 1-22/22 (END)
```

---

22. Reboot the system, then view the status of the chronyd service

```
mohammadtarek@localhost:~ — sudo systemctl status chronyd

[mohammadtarek@localhost ~]$ sudo systemctl status chronyd
[sudo] password for mohammadtarek:
○ chronyd.service - NTP client/server
     Loaded: loaded (/usr/lib/systemd/system/chronyd.service; disabled; preset:>
     Active: inactive (dead)
       Docs: man:chronyd(8)
             man:chrony.conf(5)
lines 1-5/5 (END)
```